

CFD-informed longitudinal flight dynamics simulator

Nicola Blinov | MATLAB + ANSYS Fluent | Academic project

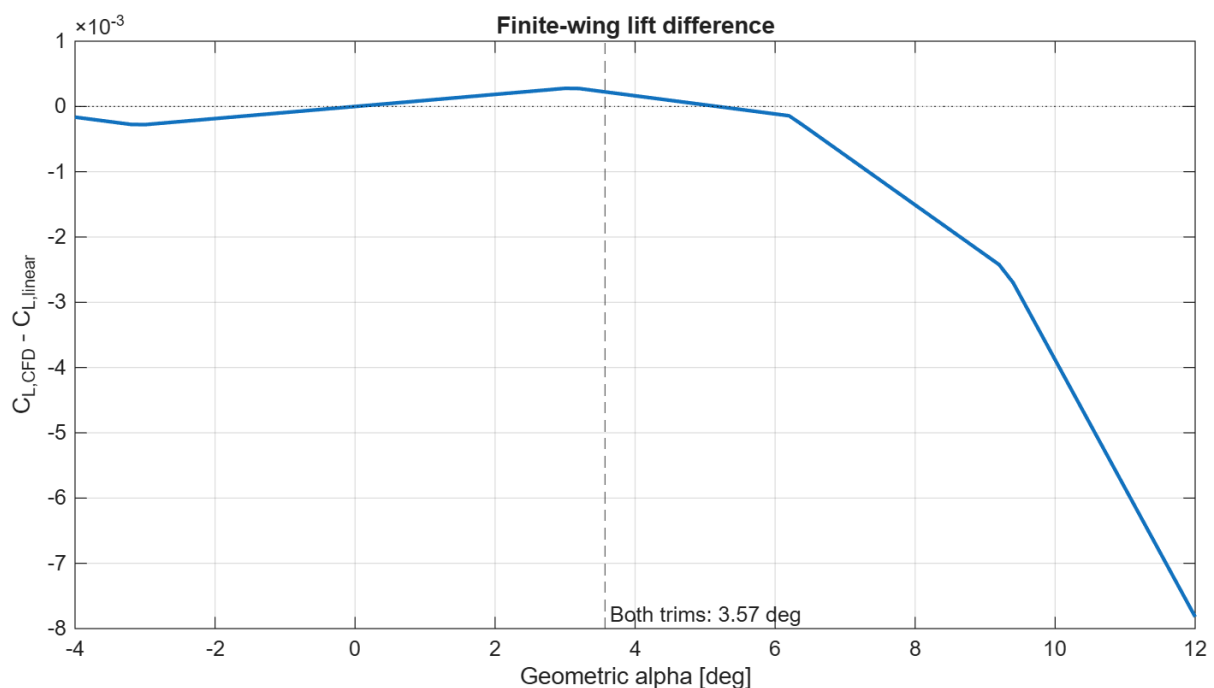
THE QUESTION

How does a measured-range CFD airfoil polar change the longitudinal response of a conceptual aircraft compared with a local linear lift law?

METHOD

NACA 0012 Fluent polar (Re 6e6, Mach 0.15, 449x129 mesh, -4 to 12 deg)
-> finite-wing correction -> separate trims -> 20 s ode45 elevator-pulse runs.

STATIC LIFT DIFFERENCE ACROSS THE AVAILABLE POLAR



TRIM, 52 M/S

3.573 / 3.570 deg

alpha: linear / CFD lift

PULSE PEAK

0.3467 / 0.3469 deg

|Delta alpha|: linear / CFD

PITCH-RATE PEAK

1.639 / 2.408 deg/s

kq = 1 / kq = 0

RESULT AND LIMITS

Near their trims, the two lift laws are nearly linear and the same -0.5 deg elevator pulse gives alpha-deviation curves within 0.000222 deg (maximum). Experimental comparison validates the airfoil polar only (CL RMSE 0.011526); the conceptual aircraft dynamics are not experimentally validated.

CONTRIBUTION

Nicola: Fluent CFD campaign, physical model setup, and part of the MATLAB code. MATLAB implementation and documentation also developed with Codex assistance.

Code, results and reproduction:

<https://github.com/nclblnv44/cfd-informed-longitudinal-flight-dynamics>

Experimental airfoil source: NASA Turbulence Modeling Resource (Ladson).